HANDS ON SUBNETING PRACTICAL

1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Given Network: 192.168.10.0/24

New Subnet Mask: 255.255.255.192 (/26)

Number of Subnets: 4

Subnet	Network Address	Subnet Mask	Broadcast Address
1	192.168.10.0/26	255.255.255.192	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.255

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

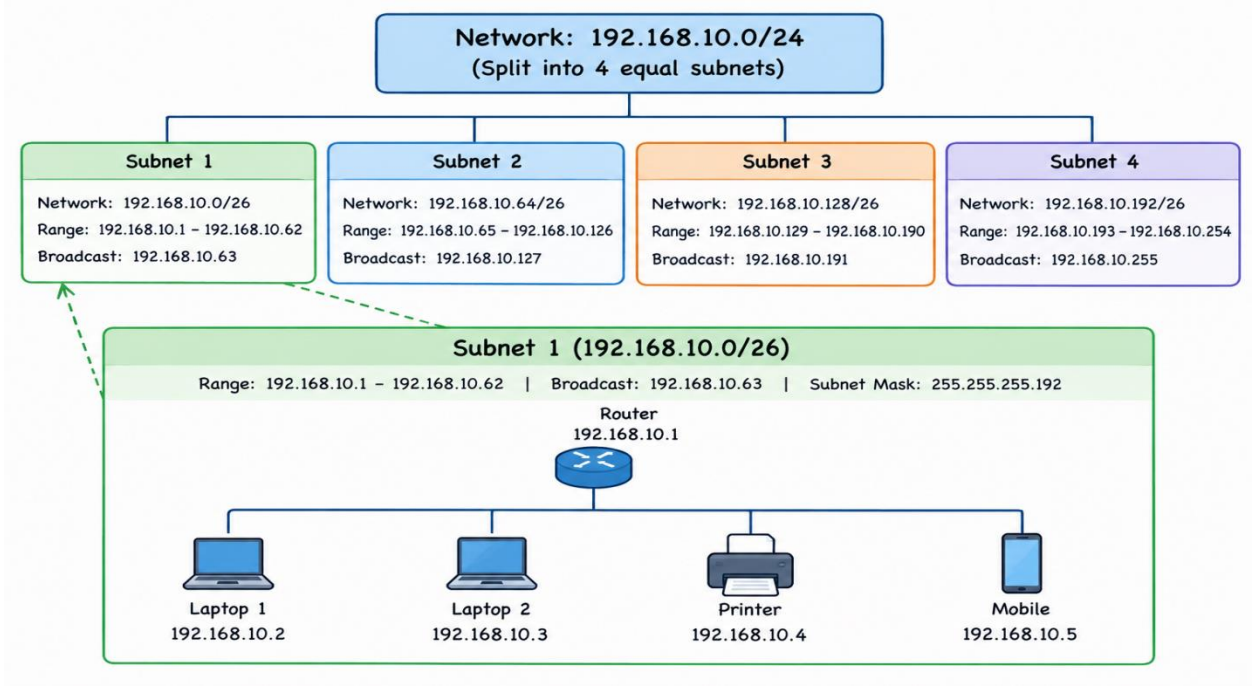
Hint: Make sure not to use the network or broadcast address for any device.

Selected Subnet: 192.168.10.0/26

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Subnet Mask: 255.255.255.192

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step-by-Step Subnetting Calculation:-

1. Given IP address: 192.168.10.0/24

2. Number of subnets: We need 4 subnets. Borrow 2 bits because \(2^2 = 4\).

3. Subnet mask: The new prefix is /26, so the subnet mask is 255.255.255.192.

4. Hosts per subnet: \(2^6 - 2 = 62\) usable host IP addresses per subnet. We subtract 2 for the network and broadcast addresses.

5. Subnet ranges:

Subnet	Network Address	Usable IP Range	Broadcast
1	192.168.10.0	.1 – .62	.63
2	192.168.10.64	.65 – .126	.127
3	192.168.10.128	.129 – .190	.191
4	192.168.10.192	.193 – .254	.255

6. Assign IP addresses: For Subnet 1, assign 192.168.10.1 to the router, .2 to PC 1, .3 to PC 2, .4 to the printer, and .5 to the server.